CAEA: Supplementary Proofs and Reproduction Details

A Proofs

A.1 Proof of Proposition 1

For any feasible π, its class marginal is b. The KL chain rule gives

$$D_{\mathrm{KL}}(\pi\|\pi_P) = D_{\mathrm{KL}}(b\|m_P) + \sum_c b(c) D_{\mathrm{KL}}\big(\pi(\cdot \mid c)\|\pi_P(\cdot \mid c)\big). \tag{1}$$

Every conditional term is nonnegative and is zero exactly when the conditional distributions agree on positive-mass classes. This yields the unique full policy π_b. Classes of zero target mass carry no response probability, so their arbitrary conditional representation does not affect uniqueness of the policy. Taking logs in Equation (1) gives $\log \pi_P = \log \pi_0 + r/\beta - \log Z$, hence $J(\pi) = \beta \log Z - \beta D_{\mathrm{KL}}(\pi\|\pi_P)$. Subtracting its value at π_P proves Equation (6). The cost is zero exactly when $b = m_P$; for full repair $b = \tau_x$. All quantities are finite by the finite response space, bounded reward, positive reference, and feasible target support. The same assumptions ensure existence of the stated optimum. □

A.2 Proof of Theorem 2

The reward difference is $r(1, s) - r(0, s) = k(2s - 1)$. A uniform reference therefore gives Equation (7). Since $\mathrm{sigmoid}(-v) = 1 - \mathrm{sigmoid}(v)$, $m_a(1 \mid 1 - s) = 1 - m_a(1 \mid s)$. A symmetric distribution of s thus has pooled report probability $1/2$. Mixture with the uniform target preserves that symmetry. Because the outcome is independent of the stance and response-sampling randomness conditional on the fixed evidence, the joint pooled law of (Z, q) is the product of uniform Bernoulli laws. Every statistic determined by that law has the same distribution, including expected Brier loss $\Pr(q \neq Z) = 1/2$ and regret $\mathbb{E}(q - 1/2)^2 = 1/4$.

Differentiating the mean-response update yields

$$F'_\lambda(s) = 1 - \eta + 2\eta(1 - \lambda)a\,\mathrm{sigmoid}(a(2s - 1))[1 - \mathrm{sigmoid}(a(2s - 1))]. \tag{2}$$

Its maximum is $1 - \eta + \eta(1 - \lambda)a/2$, attained at $s = 1/2$. The derivative is nonnegative, so a maximum strictly below one gives a global contraction. If $(1 - \lambda)a > 2$, let $d = s - 1/2$ and $g(d) = (1 - \lambda)[\mathrm{sigmoid}(2ad) - 1/2] - d$. Then $g(0) = 0$, $g'(0) > 0$, $g(1/2) < 0$, and g is strictly concave for $d > 0$. It therefore has a unique positive root and, by odd symmetry, a unique negative root. At the positive root, strict concavity gives $g'(d) < 0$, so $0 \leq F'_\lambda(s) < 1$ there. The negative root has the same derivative. The central point is unstable. Solving the strict contraction inequality proves the stated infimum, with the boundary qualification. □

A.3 Proof of Corollary 3

The upper-report probability is $\mathrm{sigmoid}(4kd(s - p)/\beta)$. Symmetry about p and $\mathrm{sigmoid}(v) + \mathrm{sigmoid}(-v) = 1$ imply a uniform pooled report law, which is independent of the stipulated outcome. The Brier identity $\mathbb{E}(Q - Z)^2 = p(1 - p) + \mathbb{E}(Q - p)^2$ gives the loss. With $v = (s - p)/d$, the repaired mean-response map becomes

$$v' = (1 - \eta)v + \eta(1 - \lambda)\tanh(cv), \qquad v \in [-1, 1].$$

Its maximum derivative is $1 - \eta + \eta(1 - \lambda)c$, at zero. When $(1 - \lambda)c > 1$, the function $(1 - \lambda)\tanh(cv) - v$ has positive derivative at zero, is negative at $v = 1$, and is strictly concave on $(0, 1]$. It has one positive root with negative derivative, and by odd symmetry one negative root; both have update derivative in $[0, 1)$. The center is unstable. These facts prove both regimes. □

A.4 Proof of Theorem 4

Couple the two repaired report kernels by first using a shared Bernoulli mixture indicator. With probability λ_x, draw the same report from the common target, giving zero report discrepancy. Otherwise use an optimal coupling of the original report kernels. Thus,

$$W_1(K^{\lambda}_{x,u}, K^{\lambda}_{x,v}) \le (1-\lambda_x) L_x |u-v|. \tag{3}$$

Apply the update to this coupling and use the triangle inequality:

$$\mathbb{E}|U' - V'| \le (1-\eta)|u-v| + \eta(1-\lambda_x)L_x|u-v| = \rho_x|u-v|. \tag{4}$$

Coupling the input belief laws and iterating proves geometric contraction when the coefficients are uniformly below one. This statement compares two evolving laws. It does not remove the within-law noise generated by independent report draws.

For approximate full repair, report values lie in an interval of diameter one, so $W_1(\widehat{\tau}_{x,s}, \tau_x) \le \mathrm{TV}(\widehat{\tau}_{x,s}, \tau_x) \le \epsilon_t$. The ideal target is independent of its current belief, permitting this coupling even when the two belief states differ. If $e_t = \mathbb{E}|U_t - \widetilde{U}_t|$, then $e_{t+1} \le (1-\eta)e_t + \eta\epsilon_t$. Unrolling from $e_0 = 0$ yields Equation (10). This bound is relative to the ideal audit-driven process. Relating that process to the evidence posterior additionally requires controlling its forecast regret, user lag, and sampling variance. □

A.5 Proof of Corollary 5

Each empirical class frequency has variance $\tau_c(1-\tau_c)/n$. Jensen's inequality followed by Cauchy–Schwarz gives

$$\mathbb{E}\,\mathrm{TV}(\bar{\tau}, \tau) \le \frac{1}{2\sqrt{n}} \sum_{c=1}^{C} \sqrt{\tau_c(1-\tau_c)} \tag{5}$$

$$\le \frac{1}{2\sqrt{n}} \sqrt{\left(\sum_c \tau_c\right)\left(\sum_c (1-\tau_c)\right)} = \frac{1}{2}\sqrt{\frac{C-1}{n}}. \tag{6}$$

Smoothing mixes the empirical target with the uniform distribution with weight $C\alpha/(n + C\alpha)$. The triangle inequality and the bound $\mathrm{TV} \le 1$ give Equation (11). □

A.6 Monotonic cost along the repair path

Let $H(\lambda) = D_{\mathrm{KL}}((1-\lambda)m_P + \lambda\tau \| m_P)$. It is convex, $H(0) = 0$, and $H(\lambda) \ge 0$. For $0 \le \lambda_1 < \lambda_2$, convexity gives $H(\lambda_1) \le (\lambda_1/\lambda_2)H(\lambda_2) \le H(\lambda_2)$. Therefore the preference objective cost is nondecreasing along the path. This does not prove optimality over all possible stable policies or all possible repair directions. □

A.7 Proof of Proposition 6

Because A is finite, choose $v \in (1/2, 1)$ such that neither v nor $u = 1 - v$ belongs to A. Both can be made arbitrarily close to the endpoints. Choose disjoint neighborhoods around u and v that avoid A and are mirror images. Let b_v be a nonnegative smooth bump supported in the neighborhood of v, with $b_v(v) = 1$ and $b_v'(v) = 0$, and set $b_u(s) = b_v(1-s)$. Define $m(s) = 1/2 + (v - 1/2)[b_v(s) - b_u(s)]$. Then $m \in [u, v]$, $m(s) = 1/2$ on A, $m(v) = v$, $m(u) = u$, and $m'(u) = m'(v) = 0$. The two corresponding fixed points have update derivative $1 - \eta$, hence are stable for $0 < \eta \le 1$. This is also realizable as a finite-temperature Gibbs policy with a uniform reference by taking $r(1, s) - r(0, s) = \beta \log[m(s)/(1 - m(s))]$, a bounded smooth reward difference.

For the certificate, Hoeffding's inequality and a union bound give $\max_j |\widehat{m}_j - m(s_j)| \le \epsilon$ with probability at least $1 - \delta$. The true secant slope on an interval differs from its empirical secant by at most $2\epsilon/h$. On an interval $[a, b]$ of length h, let $D = h^{-1}\int_a^b m'(t)\,dt$. For every $s \in [a, b]$,

$$|m'(s) - D| \le \frac{H}{h}\int_a^b |s - t|\,dt = \frac{H((s-a)^2 + (b-s)^2)}{2h} \le \frac{Hh}{2}.$$

Taking absolute values and the maximum over intervals proves Equation (12). For binary reports,

$$W_1(\mathrm{Bern}(a), \mathrm{Bern}(b)) = |a - b|.$$

With $n \simeq Nh$, the sampling term is of order $\sqrt{\log(J/\delta)/(Nh^3)}$; balancing against Hh yields the stated order, ignoring logarithms. □

B Stochastic noise floor and measurement safeguards

Under full binary repair, Q_t are independent Bernoulli(1/2) draws and $U_{t+1} = (1-\eta)U_t + \eta Q_t$. Independence gives

$$\mathrm{Var}(U_T) = (1-\eta)^{2T}\,\mathrm{Var}(U_0) + \frac{\eta}{4(2-\eta)}[1 - (1-\eta)^{2T}]. \tag{7}$$

Thus even an exactly invariant report policy can raise variance from an initial population concentrated near 1/2. Equation (14) removes this particular baseline diffusion; interpreting raw variance growth as sycophancy would be incorrect. The comparison still assumes a meaningful, matched counterfactual process and does not identify all cognitive harms.

Neither BPD nor PCD is an accuracy score. BPD can be small under a common wrong consensus, and a chosen target can itself be wrong. ECR is a proper-score regret difference, not a causal estimate of human impact. Only their joint use, with explicitly defined evidence, response interpretation, and baseline, supports the limited simulator conclusions in this paper.

C Reproduction and numerical verification

The main runner is `code/experiment.py`; stress experiments are `code/stress.py`; the optional neural component is `code/neural_probe.py`. Finite response-audit certificates use `code/audit_stability.py`, and option permutations use `code/label_control.py`. Raw per-seed CSV files, optimizer diagnostics, seeds, package versions, neural model revisions, prompt records, fitted readout weights, and plotting code accompany the manuscript. The reported runs use public model weights and computationally generated data. Model weights are fetched from the original public repositories and are not redistributed in the package.

Twenty separate 12-response, four-class constrained problems were solved with SLSQP independently of the closed-form repair. The largest difference between the numerical and analytic objective values was 2.59×10^{-13}. This checks the projection implementation rather than its applicability to natural language. The added `code/review_audit.py` independently replays archived neural scores and exports per-prompt metrics, crossed context/seed bootstrap intervals, and numerical identity checks. Inference caches now require matching model revision and prompt fingerprints; legacy unbound caches are reserved for explicit archival replay. Empty target-supported response classes raise an error instead of silently losing mass. Confidence intervals in the main table are computed over independent synthetic runs; they do not describe uncertainty across all possible preference distributions or human populations.

D Archived neural results and uncertainty scope

Table 1: Frozen-backbone results. Mean ± 95% Student-t half-width over five label seeds. These are conditional on the shared fixed prompt split and exclude prompt-population uncertainty.

Frozen backbone	Selector	Report regret	Style match	Target TV	KL to head-DPO
SmolLM2-135M	LM-base	0.1802 ± 0.0000	0.5017 ± 0.0000	0.5908 ± 0.0000	1.3573 ± 0.6896
	Head-DPO	0.1638 ± 0.0196	0.6706 ± 0.0679	0.6296 ± 0.0283	0.0000 ± 0.0000
	Audit-only	0.0145 ± 0.0000	0.5000 ± 0.0000	0.0000 ± 0.0000	2.3059 ± 0.6173
	CAEA-0.5	0.0892 ± 0.0098	0.6556 ± 0.0566	0.3148 ± 0.0141	0.6636 ± 0.2227
	CAEA-1	0.0145 ± 0.0000	0.6406 ± 0.0503	0.0000 ± 0.0000	1.9784 ± 0.4995
Qwen2.5-0.5B	LM-base	0.2389 ± 0.0000	0.5757 ± 0.0000	0.6104 ± 0.0000	1.8808 ± 0.0817
	Head-DPO	0.2044 ± 0.0137	0.7104 ± 0.0685	0.6483 ± 0.0234	0.0000 ± 0.0000
	Audit-only	0.0145 ± 0.0000	0.5000 ± 0.0000	0.0000 ± 0.0000	2.4866 ± 0.2371
	CAEA-0.5	0.1094 ± 0.0068	0.6873 ± 0.0688	0.3242 ± 0.0117	0.6654 ± 0.0937
	CAEA-1	0.0145 ± 0.0000	0.6643 ± 0.0723	0.0000 ± 0.0000	2.0261 ± 0.1859

The main paper additionally reports exploratory paired context/seed bootstrap intervals. The fixed prompt grid is not a representative sample of natural conversations. Archived logits and hidden states are replayed without fresh backbone inference. The accompanying code contains per-prompt metrics, 24 evidence clusters, five label seeds, and all 5,000 bootstrap resamples' reported quantiles. Fresh inference requires online access to the pinned public model revisions; cache fingerprints now reject unbound legacy caches in the inference runner. Explicit archival replay is available offline.